

Swesan Pathmanathan

Canadian Citizen

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EDUCATION

McMaster University

Bachelor of Engineering, Software Engineering (Co-op)

Hamilton, ON

Sep 2021 – Apr 2026

TECHNICAL SKILLS

Languages: Python, C

Systems: Linux, Bash, Windows, MAAS, ROCm

GPU Platform Validation: RAS, IO, Performance, System Management

Data: NumPy, scikit-learn

WORK EXPERIENCE

Advanced Micro Devices (AMD) | Python, Linux, MAAS, Bash, Windows

Markham, ON

Datacenter GPU Infrastructure & Validation Engineer (Co-op)

May 2024 – Aug 2025

- Debugged **MAAS** provisioning flows for MI300X, MI325X, and MI350X systems, resolving commissioning and imaging issues so GPU validation racks stayed ready for recurring automated integration runs.
- Operated automated post-silicon validation on **Linux** datacenter GPU platforms for MI300X, MI325X, and MI350X against validated BKC's, running integration passes across Performance, IO, and platform quality at 100+ executions per week.
- Deployed and maintained **Linux** and Windows GPU lab configurations with varied BMC, BIOS, and OS images to stress RAS, IO, Performance, and System Management, reducing early bring-up issues by about 20%.
- Built **Python** tooling and an internal workload launcher that accelerated regression execution, cutting recurring manual setup from about 20 minutes to under 5 minutes for validation engineers.
- Triaged hardware and software failures across ROCm, firmware, drivers, and interfaces with systematic log review and reproducible test flows, shortening time-to-root-cause on assigned issues by roughly 30%.
- Partnered with firmware and validation engineers across Jira and shared setup guides so teammates could repeat complex rack-style platform bring-up reliably in a distributed team environment.

PROJECTS

ALU Design with Sequential Circuits | Verilog, Quartus

September 2025

- Built an **Arithmetic Logic Unit (ALU)** in **Verilog** using simple, synthesizable modules, passing all planned functional tests.
- Used **Quartus** waveform views and testbenches to check that arithmetic and logic operations behaved as expected across dozens of input cases.

SufSort Sorting System | C, Bash, Linux

November 2025

- Wrote two sorting programs in **C** to compare how different algorithms behave on large data files of up to several hundred megabytes.
- Used **Bash** scripts on **Linux** to generate input data, run batch tests, and check sorted output, catching logic errors early in development.

Linux Log Corpus Analysis | Python, scikit-learn, NumPy

February 2026

- Built a simple analysis pipeline in **Python** to read and process tens of thousands of labeled Linux system logs.
- Used **NumPy** and **scikit-learn** to find which log terms best separate different log categories, improving signal-to-noise for downstream analysis.

RELEVANT COURSES

Computer Networks Security, Software Testing, Computer Performance Analysis